

Project Idea

Title: LPG Cylinder Monitoring and Gas Leakage Detection System

Description: This Arduino-based system monitors LPG usage and enhances household safety by combining gas level tracking and leakage detection. The system measures the cylinder's weight to estimate the remaining gas and alerts users before depletion. It also continuously detects gas leakage using a gas sensor. If a gas leak is detected, the system not only alerts the user but also turns on an exhaust fan to prevent accidents.

Features:

- **Real-Time Gas Level Monitoring:**
 - Load cell + HX711 measures cylinder weight
 - Displays remaining gas on LCD
- **Low Gas Warning System:**
 - Buzzer alerts when gas is low
 - LCD shows warning message
- **Gas Leakage Detection:**
 - MQ-2 Gas Sensor monitors LPG concentration
 - Detects unsafe gas levels
- **Emergency Safety Features:**
 - Exhaust Fan Activation
 - Emergency Alarm System (Buzzer & LCD Display)

Components:

- Arduino Uno R3
- Load Cell Weight Sensor (10kg TAL220)
- HX711 Load Cell Amplifier Module
- MQ-2 Gas Sensor
- LCD Display (I2C 20x4)
- Buzzer (Active)
- 5V Cooling Fan (40 x 40 x 10mm)
- 5V Relay Module for Arduino
- Adapter DC 5V 2A 2.1mm Barrel Jack Adapter
- DC Barrel Power Jack 5.5x2.1mm Female Adapter
- Jumper wires (M-F, M-M)
- Breadboard
- Lighter (Simulating Gas)
- Container (Simulating Cylinder)

Members:

- Plaban Basu - 22101831 - plaban.basu@g.bracu.ac.bd
- Abdullah Al Mahamud - 22101811- abdullah.al.mahamud@g.bracu.ac.bd
- Bishan Paul - 21201714 - bishan.paul@g.bracu.ac.bd
- Torsha Bashar Chamak - 21301083 - torsha.bashar.chamak@g.bracu.ac.bd
- Tamjeed Abdullah Noor - 19201062 - tamjeed.abdullah.noor@g.bracu.ac.bd